Differential Equations

Notes by Arjun Maneesh Agarwal

based on course by Aditya Karnataki

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1. Aditya Karnataki's Wisdom

Good to see all of you here.

TODO. *Karnataki's comments on view, rooms and property and why he has a 6th floor office.*

Good to see you all on time, <looking at the clock which is wrong>. The clock is not on time. Let me fix it, once and for all. <fixing the clock>. Abh apna sahi time chal raha hai.

We will try to include everybody, mathmeatically and non-mathematically.

Kids at CMI are more interested in watching paint dry than solving Differential Equations. I was one of them, my prof was really, really terrible. The only thing I learned in that course was how to pass the course.

All chalks gather at four places. Eventually, all chalks will gather their. This is called a sink. We will see this chalk in motion now.

The variable with respect to which we differentiate will be t. I don't differentiate with x's (ex's). I don't differentiate between my ex's. They are variable's with variable values (and costs). They are real, but sometimes complex.

There is t(tea) inside me and I liked it.

Always to lessen you headache, by making good choices. This is life gyaan, not only malthmeatical gyaan.

And that is a topic for other time. <Adie looks at the clock>. No need to look at the clock, I am ending the class.

I get off my curly $\mathcal{X}'s$(ex's). There is no pun there.[1]

Any comments? Any comments about my x? <To Adie> I know your name is also Aditya but that doesn't give you the right to comment about my x. Only I have that right. And the left.

One should be careful. Being extra careful doesn't hurt. Maybe sometimes it does. Like when I am writing my papers, I feel I am being a little too careful. I should let my intuition carry the boat.

I had to leave my fellowship for a while, but now I am back in white (he was wearing the Galois in Action TShirt), like Gandalf. I had to fight some Balrogs but now I am back.

As you might remember, I fixed the clock on day 1. Perhaps fixing your clocks is the harder part.

There is a complex analogue of the story, that is no more complex than the real part.

If something is nilpotent, I can compute it in finite time so I have no problems. In that regard, outside I have 99 but this ain't one of them. My 50 Cents for you.

<Jumping up and down and moving linearly, comparative to a frog> I can just do $x + \frac{x^2}{2} + \frac{x^3}{3!} + ...$ and it has to terminate by $2n$.

But we would like to have a bound on N, I don't want to do say 20000 computations manually. <me: Won't you use a computer for that>, In that case, million cross million matrices. Arey, point ko samjho. There has to always be a hard bound. That's why I carry a hard bound. <taps his notebook>

This answers Aditya's question. Your Aditya I mean. I have many question which no one can answers.

[1] Lesson is to not make ex jokes in algebra class. Too much risk of pun not intended.

The center is hard to understand. It's likes coming close but never converging. It's like friendzone. A lot of x's have been friendzoned.

(Adie asks for the quiz at the begining of the class) Have you never been to my class before? Or was it your evil twin? <He could be the evil twin> Maybe they both are evil. But given this twin is coming to Differential Equations, I don't think he is evil. <Engineers like DEQN.> Hmm, They are differential in the way they do things.

It's going to be a beatiful quiz. A big beatiful quiz. Nobody makes quizzes like me.

(Negi keeps his phone on the board, pre-emptivly for the quiz) I have had a close association to CMI but this is a first. Now I have a story to tell, I once had a batch that had a bunch of weirdos. No, no. I should not say this. CMI is the place for all of you. I have no issues, can't say that for other professors.

2. Tuesday 13 Jan, 2026

2.1. Simplest Differential Equation

$$x' = ax$$

Here $x(t)$ us an unknown function. We normally deal with real valued functions of real variable t.

x' represents $\frac{\mathrm{d}(x)}{\mathrm{d}(t)}$. a is a parameter.

We know $x(t) = ke^{at}$ is a solution, for any $k \in \mathbb{R}$. This is by

$$\int \frac{\mathrm{d}x}{x} = \int a \,\mathrm{d}t$$

But we want to have all solutions.

Claim 2.1. All solutions are of the form $x(t) = ke^{at}$.

Proof. If $u(t)$ is any solution, then

$$\begin{aligned} &\frac{\mathrm{d}}{\mathrm{d}t}(u(t)e^{-at} \\ &= u'(t)e^{-at} - au(t)e^{-at} \\ &= e^{-at}(u'(t) - au(t)) \\ &= 0 \end{aligned}$$

Thus, $(u(t)e^{-at}$ has derivative 0 and is hence constant. Thus, $u(x)$ has to be of the form ke^{at}. ■

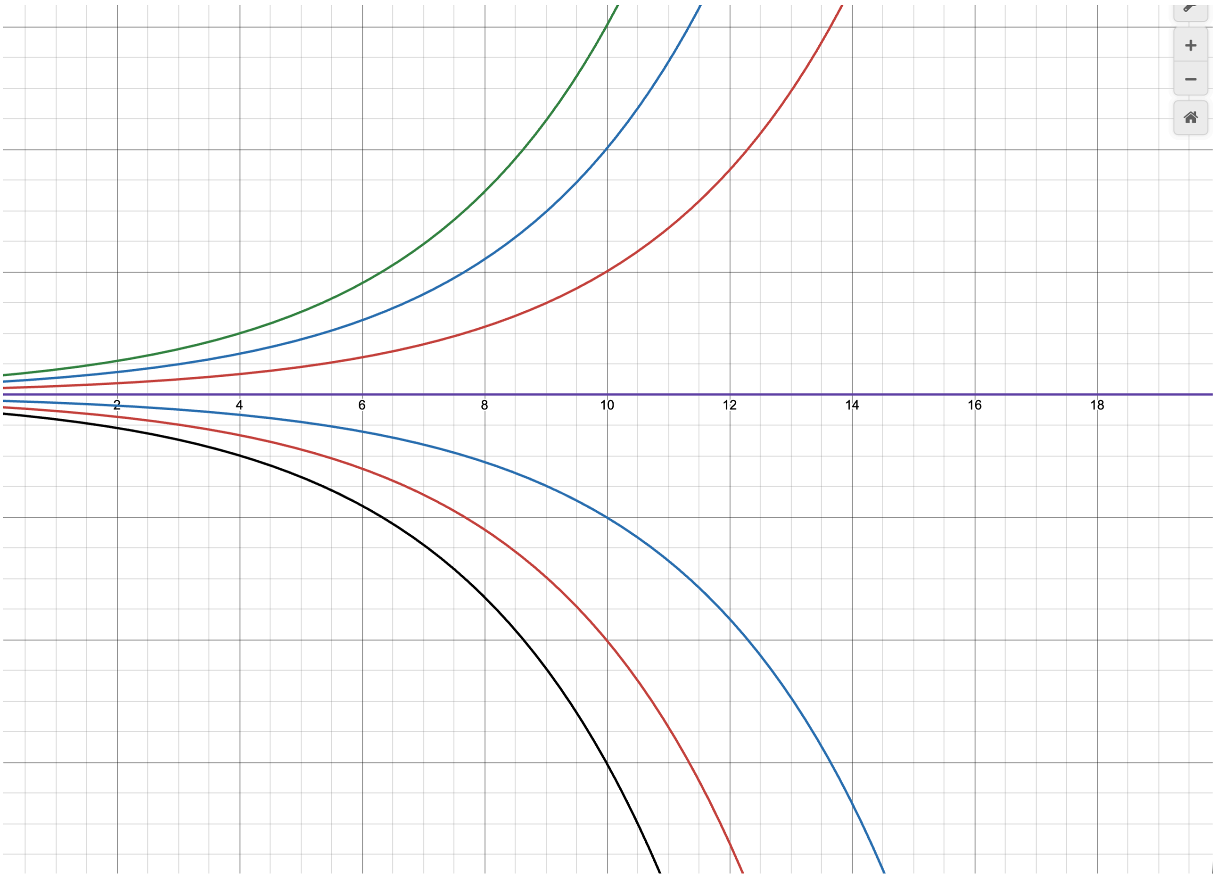
But how do we resolve k? Simple, notice that $k = u(0)$ and hence, $\boxed{u(t) = u(0)e^{at}}$. This is called the self contained form and in general we want that.

In fact $k = u(t_0)e^{-at_0}$. If $v(t) = u(t - t_0) \Rightarrow v(t_0) = u(0)$ (this is called the translational invariance), so we can assume "initial condition" to be at $t > 0$.

Side Note

When $k = 0$, $x(t) \equiv 0$. Such initial conditions are called "Equilibrium Solution's" or "Equilibrium Points".

Definition 2.2. Initial conditions such that the solution is $x(t) \equiv B$ identically are called Equilibrium Solutions or Equilibrium Points.



Here the purple line is our Equilibrium Point.

Source and Sink 2.3. If other curves diverge away from the Equilibrium point, we call it a source. If other curves had converged to this point, we would call it a sink.

2.2. Population Dynamics

Differential Equations are used to model population growth. While our simple equation works for bacteria where there is no resource bound on their population. The same can't be said about mammals. There are bounds on how much the environment can support. Consider

$$x' = ax\left(1 - \frac{x}{N}\right)$$
$$a, N > 0$$

Remark

- When population is small, rate is (approx) proportional to the size of population.
- Population exceeds N, then rate is negative.

To make our lives easier, we will assume $N = 1$ by taking x normalized as the percentage of 'ideal' population at time t.

This gives us:

$$x' = ax(1-x) = f_a(x)$$

Here f_a has the subscript to show dependence on a.

This is the first example of:

$$\underbrace{\text{First Order}}_{\substack{\text{Only 1st}\\\text{derivative}}}, \underbrace{\text{autonomous}}_{\substack{\text{RHS depends on x,}\\\text{but not explicitly on t}}}, \underbrace{\text{non-linear}}_{\substack{\text{RHS is}\\\text{not linear}}} \text{ differential equation}$$

Solution.

$$\int \frac{\mathrm{d}x}{x(1-x)} = \int a\,\mathrm{d}t$$

$$\int \frac{1}{x}\,\mathrm{d}x + \int \frac{1}{1-x}\,\mathrm{d}x = \int a\,\mathrm{d}t$$

$$x(t) = \frac{Ke^{at}}{1+Ke^{at}}$$

where K is constant of integration. Notice

$$K = \frac{x(0)}{1-x(0)}$$

Thus, we can make a self contained form

$$\boxed{x(t) = \frac{x(0)e^{at}}{1-x(0)+x(0)e^{at}}}$$

■

Notice, $x(t) \equiv 1$ and $x(t) \equiv 0$ are both equilibrium solutions.

TODO. *Drawing of sink and source*

Notice, $x(t) \equiv 0$ is a source and $x(t) \equiv 1$ is a sink.

Notice, we can get all this via the slope field as well. A solution just follows one of the sets of arrows in the slope field.

TODO. *Slope Field*

2.2.1. Graph of $f_a(x), ax(1-x)$

The roots $x = 0, x = 1$; they are equilibrium points.

It is easy to compute that $f_a'(x) = a - 2x$. Notice,

$$f_a'(0) = a > 0$$
$$f_a'(1) = -a < 0$$

This is an analogue of derivative test and works on autonomous system.

We could add time dependence to the RHS to model harvesting or selling off etc. But that is something we'll look at later. Let's move to some variable based math.

2.3. System of Differential Equations

Consider

$$\begin{aligned} x_1' &= f_1(t, x_1, ..., x_n) \\ x_2' &= f_1(t, x_1, ..., x_n) \\ &\vdots \\ x_n' &= f_1(t, x_1, ..., x_n) \end{aligned}$$

Assume[2] $f_i \in C^\infty$ that is partial derivatives of all orders of every f_i exists and are continuous.

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

We can write this all as

$$X' = F(t, X) = \begin{pmatrix} f_1(t, x_1, ..., x_n) \\ f_2(t, x_1, ..., x_n) \\ \vdots \\ f_n(t, x_1, ..., x_n) \end{pmatrix}$$

A more general way could be:

$$X'(t) = F(t, X(t))$$

Unfortunately, there is no guarantee that a solution exists.

One major theorem we will do close to the midsem will be that the solution always locally exists over t. The problem will be if we can stitch them together.

Redefining some stuff,

Definition 2.4. An equation is autonomous if $F(t, X) = F(X)$

[2]For the sake of your sanity and mine

Definition 2.5. X_0 such that $F(X_0) \equiv 0$ is equilibrium point coresponds to $X(t) \equiv X_0$

Remark: Motivation

The motivation for this comes from Physics or second order differential equations.

$$mx'' = f(x)$$
$$mx'' + bx' + kx = 0$$

The first comes from an obscure guy named Newton while the second is a Damped Harmonic Oscillation.

We can write these as a system of equations.

Let $y = x'$,

$$\begin{aligned} x'' + ax' + bx = 0, y = x' \Rightarrow x' &= y \\ y' &= -bx - ay \end{aligned}$$

We can write

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -b & -a \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

We will see how we can solve these later. In the last two minutes of this class, I will go for a simpler example.

2.4. Uncoupled Linear System

Consider

$$X' = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix} X;$$
$$\Rightarrow X(t) = \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{2t} \end{pmatrix} X(0)$$

This works in diagonal matrices. But what about arbitrary matrices? We will see it next time.[3]

3. Friday 16 Jan, 2026

3.1. Recap

Consider

[3] Prof. Karnataki was interrupted by Prof. Madhavan...

$$X' = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix} X;$$
$$\Rightarrow X(t) = \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{2t} \end{pmatrix} X(0)$$

As function of x_1, x_2, solution curves llie on $y = \frac{k}{x^2}$ where $k = c_1^2 c_2$.

3.2. Non Diagonal Matrices

Recall If $\lambda_1, \lambda_2, ..., \lambda_n$ are real distinct eigenvalues of $n \times n$ matrix A, then there exists an invertible $n \times n$ matrix such that

$$P^{-1}AP = \operatorname{diag}(\lambda_1, \lambda_2, ..., \lambda_n)$$

If $X' = AX$, define $Y = P^{-1}X$. Then

$$Y' = P^{-1}X' = P^{-1}AX = (P^{-1}AP)(P^{-1}X)$$
$$\Rightarrow Y' = (P^{-1}AP)Y$$
$$\Rightarrow Y(t) = \operatorname{diag}\left(e^{\lambda_1 t}, ..., e^{\lambda_n t}\right)Y(0) = E(t)Y(0)$$
$$\Rightarrow X(t) = PE(t)P^{-1}X(0)$$

Lemma 3.1. *$X' = AX$. Suppose V_0 is an eigenvector for A with eigenvalue $\lambda \in \mathbb{R}$. Then $x(t) = e^{\lambda t}V_0$ is a solution to the system.*

Proof. This is because $X'(t) = \lambda e^{\lambda t}V_0 = e^{\lambda t}(\lambda V_0) = e^{\lambda t}(AV_0) = AX(t)$ ■

Example

Consider $x_1' = -x_1 - 3x_2$ and $x_2' = 2x_2$. This gives

$$A = \begin{pmatrix} -1 & -3 \\ 0 & 2 \end{pmatrix}$$

This gives the eigenvalues $\lambda_1 = -1, \lambda_2 = 2$ and the corresponding eigenvectors $V_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, V_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. Furthermore,

$$P = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \Rightarrow P^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Remark

Something worth remembering is that upper triangular matrices with diagonal being equal are inverted by just taking negative of all off diagonal elements.

$$P^{-1}AP = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix} \Rightarrow y_1' = -y_1, y_2' = 2y_2$$

Then,

$$x(t) = P\begin{pmatrix} e^{-t} & 0 \\ 0 & e^{2t} \end{pmatrix} P^{-1} X(0)$$
$$\Rightarrow x_1(t) = c_1 e^{-t} + c_2(e^{-t} - e^{2t}); x_2(t) = c_2 e^{2t}$$

3.3. Phase Potraits

TODO. *Phase Potraits*

Equilibrium Definition 3.2. If $\det A \neq 0$, $(0, 0, ..., 0)$ is the only equilibrium solution. If $\det A = 0$, they will correspond to the kernel of A

Definition 3.3. If A has negative eigenvalue $\lambda_1, \lambda_2, ...\lambda_n$. and positive eigenvalue $\lambda_{k+1}, ..., \lambda_n$ all distinct. Let $\{v_1, v_2, ..., v_n\}$ be the corresponding eigenvectors. Then stable subspace refers to:

$$E^{\text{s}} = \text{span}\ \{v_1, ..., v_k\}$$
$$E^{\text{un}} = \text{span}\ \{v_{k+1}, ..., v_n\}$$

3.4. Matrix Exponentials

We need convergence of $L(\mathbb{R}^n)$ = subspace of linear operators $\mathbb{R}^n \longrightarrow \mathbb{R}^n$

Recall that **Operator Norm**: $T \in L(\mathbb{R}^n), \|T\| := \max_{|x|\ \leq 1} |T(x)|$

Lemma 3.4. *For $S, T \in L(\mathbb{R}^n)$, $x \in \mathbb{R}^n$.*

(a) $\|T\| > 0$ and $\|T\ \| = 0 \Longleftrightarrow T \equiv 0$

(b) $\|T\| = |k|\ \|T\|$ for $k \in \mathbb{R}$

(c) $\|S + T\| \leq \|S\| + \|T\|$

(d) $\|T(x)\| \leq \|T\|\ |x|$

(e) $\|TS\| \leq \|T\|\ \|S\|$

(f) $\|T^k\| \leq \|T\|^k$ for $k \in \mathbb{N}$

Definition 3.5. $T_k \longrightarrow T$ if $\forall \varepsilon > 0, \exists N$ s.t.

$$\|T_k - T\| < \varepsilon \forall k \geq n$$

Theorem 3.6. *Given $T \in L(\mathbb{R}^n)$ and $t_0 > 0$,*

$$\sum_{k=0}^{\infty} \frac{T^k t^k}{k!} \text{ converges absolutely and uniformly } \forall\ |t| \leq t_0$$

Proof. Let $\|T\| = a$

$$\| \frac{T^k t^k}{k!} \| \leq \frac{\|T\|^k \ |t|^k}{k!} = \frac{a^k \ |t|^k}{k!}$$

Since $\sum_{k=0}^{\infty} \frac{a^k t_0^k}{k!} = e^{a t_0}$, it follows from M test that it indeed coverges for $|t| < t_0$. ∎

Definition 3.7.

$$e^T := \sum_{k=0}^{\infty} \frac{T^k}{k!} \in L(\mathbb{R}^n)$$

Notice, $\|e^T\| \leq e^{\|T\|}$

Definition 3.8. For a matrix A,

$$e^{At} = \sum_{k=0}^{\infty} \frac{A^k t^k}{k!} \in L(\mathbb{R}^n)$$

Notice, $\|e^{At}\| \leq e^{\|At\|}$

Proposition 3.9. *If $S = PTP^{-1}$, then*

$$e^S = Pe^T P^{-1}$$

Corollary 3.10. *If $P^{-1}AP = \text{diag }(\lambda_1, \lambda_2, ..., \lambda_n)$ then $e^{At} = Pe^{\text{diag }(\lambda_1, \lambda_2, ..., \lambda_n)} P^{-1}$*

Proposition 3.11. *If $S, T \in L(\mathbb{R}^n)$ that commute, then $e^{S+T} = e^S e^T$*

Proof. If $ST = TS$, then by binomial theorem,

$$(S+T)^n = n! \sum_{j+k=n} \frac{S^i T^k}{j!k!}$$

$$\therefore e^{S+T} = \sum_{n=0}^{\infty} \sum_{j+k=n} \frac{S^j T^k}{j!k!} = e^S e^T$$

∎

Corollary 3.12. $\left(e^T\right)^{-1} = e^{-T}$

Corollary 3.13. *If $A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$, $e^A = e^a \begin{pmatrix} \cos b & -\sin b \\ \sin b & \cos b \end{pmatrix}$*

Proof. Let $\lambda = a + bi$, write A^k in terms of $\mathfrak{R}(\lambda^k)$ and $\mathfrak{I}(\lambda^k)$ ■

Corollary 3.14. *If* $A = \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix}$, $e^A = e^a \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$

Proof. $A = aI + b$ where $B = \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix}$.

Then aI and B commute

$$e^A = e^{aI} e^b = e^a e^B = e^a (I + B + 0 + \ldots) = e^a \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$$

■

For any invertible 2×2 matrix A, matrix of $e^{At} = P e^{Bt} P^{-1}$ where

$$e^{Bt} = \begin{cases} \begin{pmatrix} e^{\lambda t} & 0 \\ 0 & e^{\mu t} \end{pmatrix} \\ e^{\lambda t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \\ e^{@} \begin{pmatrix} \cos bt & -\sin bt \\ \sin bt & \cos bt \end{pmatrix} \end{cases}$$

4. Tuesday 27 Jan, 2026

Side Note

Prof. Karnataki is away attending a very important conference in Mumbai, which just happened to be conveniently located near a Linkin Park concert. Even though the tickets sold out in minutes and conferences are announced much later, miracles do happen. Therefore, Prof. Priyavrat will be revising the last class.

5. Thursday 29 Jan, 2026

Side Note

Prof. Priyavrat in revision mode.

6. Tuesday 2 Feb, 2026

Side Note

TODO.

7. Thursday 5 Feb, 2026

Theorem 7.1. *Let A be a real $2n \times 2n$ matrix with complex eigenvalues $\lambda_j = a_j + ib_j$, $\overline{\lambda_j} = a_j - ib_j, j = 1, 2, \ldots, n \left(b_j = 0\right)$. Then there exists a generalized complex eigen-*

vectors $w_j = u_j + iv_j$, $\overline{w_j} = u_j - iv_j$ *such that* $P = \begin{pmatrix} v_1 u_1 & v_2 u_2 & \dots & v_n u_n \end{pmatrix}$ *is invertible and* $A = S + N$ *such that:*

$$P^{-1}SP = \operatorname{diag}\begin{pmatrix} a_j & b_{-j} \\ b_j & a_j \end{pmatrix},$$

N *is nilpotent of order* $k \leq 2n$ *and* S *and* N *commute.*

Recall 2×2 matrix A, repeated eigenvalues λ, only 1 eigenvector v_1 then in v_1, v_2 basis (v_2 is arbitrary) then matrix A in this basis is $\begin{pmatrix} \lambda & b \\ 0 & \lambda \end{pmatrix}$.

Theorem 7.2. *Let* A *be a real* $n \times n$ *matrix with real eigenvalues* $\lambda_j, j = 1, 2, \dots, k$, *complex eigenvalues* $\mu_j = a_j + ib_j, \overline{\mu_j} = a_j - ib_j, j = k+1, \dots, \frac{n+k}{2}$. *Then there exists a basis* $\left\{v_1, \dots, v_k, t_{k+1}s_{k+1}, \dots, t_{\frac{n+k}{2}}s_{\frac{n+k}{2}}\right\}$ *for* $\mathbb{R}^n$ *such that* $P = \begin{pmatrix} v_1 & \dots & v_k & t_{k+1}s_{k+1} & \dots & t_{\frac{n+k}{2}}s_{\frac{n+k}{2}} \end{pmatrix}$ *is invertible and* $P^{-1}AP = \operatorname{diag}(B_1, B_2, \dots, B_k)$ *where* B_j *for real eigenvalue* λ *is of the form:*

$$B = \begin{pmatrix} \lambda & 1 & 0 & 0 & \dots & 0 \\ & \lambda & 1 & 0 & \dots & 0 \\ & & \lambda & 1 & \dots & 0 \\ & & & \ddots & \ddots & \vdots \\ & & & & \lambda & 1 \\ & & & & & \lambda \end{pmatrix}$$

Example

For complex eigenvalue $\lambda = a + ib$,

$$B = \begin{pmatrix} D & I_2 & 0 & 0 & \dots & 0 \\ & D & I_2 & 0 & \dots & 0 \\ & & \lambda & D & \dots & 0 \\ & & & \ddots & \ddots & \vdots \\ & & & & D & I_2 \\ & & & & & D \end{pmatrix}$$

where $D = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, 0 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

$$e^{Bt} = e^{\lambda t}\begin{pmatrix} 1 & t & \frac{t^2}{2!} & \frac{t^3}{3!} & \dots & \frac{t^{m-1}}{(m-1)!} \\ & 1 & t & \frac{t^2}{2!} & \dots & \frac{t^{m-2}}{(m-2)!} \\ & & 1 & t & \dots & \frac{t^{m-3}}{(m-3)!} \\ & & & \ddots & \ddots & \vdots \\ & & & & 1 & t \\ & & & & & 1 \end{pmatrix}$$

Example

Say $N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ then $N^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

Note, N^3 will be 0 as the 1 will fly out.

Example

$$e^{Bt} = e^{at} \begin{pmatrix} R & Rt & R\frac{t^2}{2!} & R\frac{t^3}{3!} & \dots & R\frac{t^{m-1}}{(m-1)!} \\ & R & Rt & R\frac{t^2}{2!} & \dots & R\frac{t^{m-2}}{(m-2)!} \\ & & R & Rt & \dots & R\frac{t^{m-3}}{(m-3)!} \\ & & & \ddots & \ddots & \vdots \\ & & & & R & Rt \\ & & & & & R \end{pmatrix}$$

where $R = \begin{pmatrix} \cos bt & -\sin bt \\ \sin bt & \cos bt \end{pmatrix}$

7.1. Stability Theory

Let $w_j = s_j + it_j$ be (generalized) eigenvectors corresponding to $\lambda_j = a_j + ib_j$, with b_j allowed to be zero $\left(t_j = \vec{0}\right)$.

Definition 7.3.

- $E^S = \text{Span}\left\{s_j, t_j \mid a_j < 0\right\}$ is the stable subspace
- $E^C = \text{Span}\left\{s_j, t_j \mid a_j = 0\right\}$ is the center
- $E^U = \text{Span}\left\{s_j t_j \mid a_j > 0\right\}$ is the unstable subspace

Example

$$A = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

As the matrix is pre-blocked, we have the eigenvalues $\pm i, -1$

Then, $X(t) = x_0 \begin{pmatrix} \cos t \\ -\sin t \\ 0 \end{pmatrix} + y_0 \begin{pmatrix} \sin t \\ \cos t \\ 0 \end{pmatrix} + z_0 e^{-t} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

This is stable along z axis and the center is xy plane.

In this case, the phase potrait is a bunch of cylinderical helices around the z axis with a exponentially decreasing pitch. The z axis just points to 0.

Notice, solutions starting in stable subspace stays in the stable subspace all the the time.

A question we have is the above remark true for the other subspaces.

Definition 7.4. $e^{At}:\mathbb{R}^n \to \mathbb{R}^n$ is called the flow of the linear system. If all eigenvalues have non-zero real part, this is called hyperbolic flow.

Theorem 7.5. *E^S, E^U, E^C are invariant under the flow e^{At} and $\mathbb{R}^n = E^S \oplus E^U \oplus E^C$.*

8. Tuesday 10 Feb, 2026

Side Note

I had a rust quiz just next class and hence, didn't show up.

9. Thursday 12 Feb, 2026

9.1. Existence and Uniqueness

J interval in $\mathbb{R}$, $0 \in J$, $X : J \to E \subset \mathbb{R}^n$.

$X'(t) = F(X(t)), X(0) = X_0$.

Integrating, we have:

$$X(t) = X_0 + \int_0^t F(X(s))\,\mathrm{d}s$$

This seems sort of like a continuous recursive equation. We would like to somewhat substitute $X(t)$ again and again but we can't really deal with $F\left(\int X(t)\,\mathrm{d}x\right)$ type stuff.

Assumptions $\overline{B_\varepsilon}$ closed ball of radius $\varepsilon > 0$ at X_0.

F has a Lipschitz constant K on $\overline{B_\varepsilon}$.

$|F(X)| < M$ on $\overline{B_\varepsilon}$, we choose $a > 0$ such that $a < \min\left(\frac{\varepsilon}{M}, \frac{1}{K}\right)$ and let $J = [-a, a]$.

Idea : Picard Iteration

Iteratively substitute for $X(s)$

$$\begin{aligned}
u_0(t) &= X_0 && \text{For } t \in J, \text{define} \\
u_1(t) &= X_0 + \int_0^1 F(u_0(s))\,\mathrm{d}s \\
&= X_0 + \int_0^t F(X_0)\,\mathrm{d}s && \text{Since } |t| \le a, |F(X_0)| \le M, \text{it follows that} \\
&= X_0 + tF(X_0) && u_0(t) = |u_1(t) - X_0| \le aM < \varepsilon
\end{aligned}$$

Define by induction, $u_{k+1}(t) := X_0 + \int_0^t F(u_k(s))\,\mathrm{d}s$.

Claim 9.1.

$$|u_{k+1}(t) - u_k(t)| \leq (aK)^k L \text{ for some constant } L$$

Proof. Let $L = \max_{-a \leq t \leq a} |u_1(t) - u_0(t)|$.

Then $L \leq aM$.

(B) Then

$$\begin{aligned} u_2(t) - u_1(t) \mid &= \mid \int_0^t [F(u_1(s)) - F(u_0(s))] \, \mathrm{d}s \mid \\ &\leq \int_0^t K \, |u_1(s) - u_0(s)| \, \mathrm{d}s \\ &\leq aKL \end{aligned}$$

(S) Assume by induction, $|u_k - u_{k-1}| \leq (aK)^{k-1} L$

Then,

TODO.

And we are done, by induction! ∎

Notice, $\alpha := aK < 1$ then choose

TODO.

Thus, UNIFORM CONVERGENCE!

Notice, $u_k \to X$ as $k \to \infty$

Since, $u_{k+1}(t) = X_0 + \int_0^t F(u_k(s)) \, \mathrm{d}s$. Take limit $k \to \infty$

$$\begin{aligned} X(t) &= X_0 + \lim_{k \to \infty} \int_0^t F(u_k(s)) \, \mathrm{d}s \\ &= X_0 + \int_0^1 \lim_{k \to \infty} F(u_k(s)) \, \mathrm{d}s \\ &\overset{\text{uniform}}{\underset{\text{convergence}}{=}} X_0 + \int_0^t F(X(s)) \, \mathrm{d}s \end{aligned}$$

Uniqueness? Suppose X, Y are two solutions with $X(0) = Y(0) = X_0$. Let $Q := \max_{t \in J} |X(t) - Y(t)|$ attained at $t_1 \in J$ (compactness).

$$
\begin{aligned}
Q = |X(t_1) - Y(t_1)| &= |\int_0^t [X'(s) - Y'(s)]\, \mathrm{d}s| \\
&\leq \int_0^t |F(X(s)) - F(Y(s))|\, \mathrm{d}s \\
&\leq \int_0^t K \mid X(s) - Y(s)|\, \mathrm{d}s \\
&\leq aKQ
\end{aligned}
$$

But $aK < 1 \Rightarrow Q = 0$.

Example

$X' = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) X, X_0 = \left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$.

$$
u_0(t) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}
$$

,

$$
\begin{aligned}
u_1(t) &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \int_0^t F\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) \mathrm{d}s \\
&= \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \int_0^t \begin{pmatrix} 0 \\ -1 \end{pmatrix} \mathrm{d}s \\
&= \begin{pmatrix} 1 \\ -t \end{pmatrix}
\end{aligned}
$$